

STAT/MA 41600
In-Class Problem Set #18: September 30, 2026
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Problem Set 18 Answers

1a. We have $P(X > 3) = 1 - P(X = 0) - P(X = 1) - P(X = 2) - P(X = 3) = 1 - e^{-5}(5^0/0! + 5^1/1! + 5^2/2! + 5^3/3!) = 1 - 118e^{-5}/3 = 0.7350$.

1b. We have $P(X > 3 \mid X > 1) = \frac{P(X > 3 \ \& \ X > 1)}{P(X > 1)} = \frac{P(X > 3)}{P(X > 1)} = \frac{1 - P(X \leq 3)}{1 - P(X \leq 1)}$ which evaluates to $\frac{1 - e^{-5}(5^0/0! + 5^1/1! + 5^2/2! + 5^3/3!)}{1 - e^{-5}(5^0/0! + 5^1/1!)} = 0.7659$.

2. Since the X_j are independent Poisson random variables, then X is a Poisson random variable too. We can add their mean values to get $E(X) = (5)(0.7) = 3.5$. We conclude that $P(2 \leq X \leq 6) = e^{-3.5}((3.5)^2/2! + (3.5)^3/3! + (3.5)^4/4! + (3.5)^5/5! + (3.5)^6/6!) = 0.7988$.

3. The number of students with a chocolate allergy has a Binomial distribution with $n = 50,000$ and $p = \frac{1}{15,000}$. Therefore, the number of students with a chocolate allergy has a distribution that is approximately like a Poisson random variable with mean $50,000/15,000 = 10/3$. So the probability that 4 or more students on the campus have a chocolate allergy is approximately $1 - e^{-10/3}((10/3)^0/0! + (10/3)^1/1! + (10/3)^2/2! + (10/3)^3/3!) = 1 - 1301e^{-10/3}/81 = 0.4270$.

4. We have $P(Y > X) = \sum_{x=0}^{\infty} \frac{(e^{-5})(5^x)}{x!} \sum_{y=x+1}^{\infty} (4/5)^{y-1}(1/5) = \sum_{x=0}^{\infty} \frac{(e^{-5})(5^x)}{x!} (4/5)^x = \sum_{x=0}^{\infty} \frac{(e^{-5})(4^x)}{x!} = e^{-1} \sum_{x=0}^{\infty} \frac{(e^{-4})(4^x)}{x!} = e^{-1}$.

5. We have $E((X)(X-1)(X-2)) = \sum_{x=0}^{\infty} (x)(x-1)(x-2) \frac{e^{-3.5}(3.5)^x}{x!}$, and the first three terms are each 0, so this becomes $E((X)(X-1)(X-2)) = \sum_{x=3}^{\infty} (x)(x-1)(x-2) \frac{e^{-3.5}(3.5)^x}{x!} = \sum_{x=3}^{\infty} \frac{e^{-3.5}(3.5)^x}{(x-3)!} = (3.5)^3 \sum_{x=3}^{\infty} \frac{e^{-3.5}(3.5)^{x-3}}{(x-3)!} = (3.5)^3 \sum_{x=0}^{\infty} \frac{e^{-3.5}(3.5)^x}{x!} = (3.5)^3 = 343/8 = 42.875$.